Conditional Extreme-Path Benchmarks for Sequential Probability Forecasts

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Abstract

Real-time probability forecasts for binary outcomes are now routine in sports, online experimentation, medicine, finance, and reliability monitoring. Yet retrospective interpretation often turns on pathwise extremes, such as a forecast that becomes highly confident for an event that ultimately does not occur. Standard pointwise calibration tools do not quantify how often these episodes should arise under correct sequential updating. We develop conditional extreme-path benchmark laws for sequentially calibrated forecasts, focusing on the peak-on-loss: the highest forecast probability attained on trajectories that end in failure. For continuous time with continuous paths, we obtain an exact closed-form benchmark; for discrete time, we prove sharp finite-sample bounds and an explicit correction decomposition that separates terminal-step non-attainment from first-passage overshoots. These results provide model-agnostic null reference curves and one-sided tail probabilities for trajectory-level calibration diagnostics. As corollaries, we obtain competitive summaries used in win-probability feeds, including the eventual loser's peak in two-team games and the eventual winner's trough in multi-team games. A simulation study characterizes operating behavior under calibration and misspecification. An empirical illustration using ESPN win-probability series for regular-season NFL and NBA games (2018–2024) finds broad agreement in the NFL and systematic departures in the NBA.

1 Introduction

Most calibration diagnostics answer pointwise or aggregate questions. They do not benchmark *conditional pathwise extremes*, such as "the forecast reached 90% for an event that did not occur." Without such a benchmark, these episodes are hard to interpret: they may signal miscalibration, or they may be expected under correct updating.

Real-time probability trajectories are now routine in sports, online experimentation, medicine, finance, and reliability monitoring. Across these settings, the inferential object is the *full path*, not a single forecast at a fixed time. We therefore study the following methodological problem: *under correct sequential calibration, how often should extreme forecasts appear on trajectories that ultimately fail?*

Core methodological contribution. This paper introduces *conditional extreme-path benchmarks* for sequential probability forecasts and turns them into practical calibration diagnostics. Our primary object is the *peak-on-loss* law, the distribution of

$$M_N := \max_{0 \le k \le N} p_k \quad \text{conditional on } Y = 0,$$

where p_k is the ideal sequential forecast for a binary terminal outcome Y.

Sequential calibration and martingales. Let $Y \in \{0, 1\}$ be the terminal outcome revealed at time N, and let $(\mathcal{F}_k)_{k=0}^N$ denote the forecaster's information. Under ideal sequential calibration,

$$p_k \equiv \mathbb{P}(Y = 1 \mid \mathcal{F}_k), \qquad k = 0, 1, \dots, N,$$

is a bounded Doob martingale with $p_0 = \mathbb{P}(Y = 1) \in (0, 1)$ and $p_N = Y$. In applications, a reported sequence $\hat{p}_k$ estimates p_k; persistent departures from this martingale structure are a natural notion of sequential miscalibration (Dawid, 1982; Foster and Vohra, 1998; Gneiting et al., 2007).

Why conditional extrema are the right target. Classical maximal inequalities control unconditional events such as $\{\sup_t p_t \ge x\}$, and pointwise calibration tools assess reliability at fixed times. Neither answers the retrospective question of interest: how large can peaks be on the subset of paths that end in failure? Conditioning on the terminal outcome aligns the null with that event.

What we contribute. We derive closed-form conditional benchmark laws for the peak-on-loss object and show how to use them as practical diagnostics.

- **Core law.** For binary-outcome forecasts, we derive the conditional law of $M = \sup_t p_t$ given $Y = 0$ in continuous time (Theorem 1) and sharp discrete-time bounds with an explicit correction decomposition (Theorem 2, Remark 1).
- **Diagnostics.** We convert these laws into null curves, one-sided tail probabilities, and PIT-based aggregate tests for forecast paths (Section 4).
- **Competitive corollaries.** The eventual loser's peak in two-outcome contests and the eventual winner's trough in n-outcome contests are corollaries of the same construction (Section 5).
- **Simulation operating profile.** A compact simulation block quantifies discrete-time conservatism and shows directional PIT signatures under overreaction, drift, smoothing, and reporting artifacts (Section 6).
- **Illustration.** Applied to ESPN win-probability series for NFL and NBA games (2018–2024), the diagnostics show broad agreement in the NFL and clear departures in the NBA (Section 7).

Organization. Section 2 reviews related work. Section 3 presents the main results (with proofs, derivations, and supplementary benchmarks in the Appendix). Section 4 summarizes the implied diagnostics and tail probabilities. Sections 5 and 6 cover competitive corollaries and simulation operating characteristics. Section 7 presents the NFL/NBA application. Section 8 concludes.

2 Related Work

Calibration and forecast evaluation. Classical evaluation emphasizes reliability and sharpness, often via proper scoring rules. The Brier score and its decompositions (Brier, 1950; Murphy, 1973), decision-theoretic calibration/refinement (DeGroot and Fienberg, 1983), and modern syntheses (Dawid, 1982; Gneiting et al., 2007) all focus primarily on fixed-time or aggregate behavior. Our target differs: conditional extremes of the forecast path.

Sequential forecasting and the prequential viewpoint. For time-evolving forecasts, the natural ideal is the Doob martingale $p_k = \mathbb{P}(Y = 1 \mid \mathcal{F}_k)$, so evaluation is inherently path-dependent. The prequential perspective (Dawid, 1984) treats the predictive sequence itself as the object of criticism, and online-learning work studies asymptotic calibration under weak assumptions (Foster and Vohra, 1998). We instead provide nonasymptotic distributional laws for path extremes.

Anytime-valid inference and sequential calibration tests. Recent work on anytime-valid inference develops nonnegative (super)martingales and e-values/e-processes for optional stopping and sequential testing (e.g., Howard et al., 2020; Ramdas et al., 2023), including sequentially valid approaches to forecast calibration (Arnold et al., 2023). These methods provide sequential testing procedures. Our contribution is complementary: explicit reference distributions for extreme-path statistics, with closed-form tails that are exact in a continuous-path limit and conservative in discrete time.

Martingale extrema and distributional benchmarks. Classical maximal inequalities (Doob, Ville) give sharp time-uniform *unconditional* bounds on events such as $\{\sup_t p_t \geq x\}$. Our focus is the corresponding *conditional* question. In continuous time, exact identities for suprema are known in specific settings (Nikeghbali and Yor, 2006), and related constructions characterize feasible joint laws of extrema and terminal values. We specialize to bounded Doob martingales with binary terminal value $p_N \in \{0, 1\}$ and obtain usable conditional laws (and sharp discrete-time bounds).

Methodological gap. Pointwise calibration tools identify fixed-time reliability distortions but do not benchmark conditional path extremes. Sequential calibration tests and anytime-valid methods provide time-uniform evidence against calibration, but not closed-form conditional reference curves for extreme path events. Classical martingale maximal inequalities provide sharp unconditional control of $\mathbb{P}(\sup_t p_t \geq x)$, but they do not condition on terminal failure. Our framework fills that gap with closed-form (or conservative discrete-time) conditional extreme-path laws, trajectory-level tail probabilities, and PIT diagnostics for high-confidence trajectories that end in failure.

3 Model and Main Results

Let $Y \in \{0, 1\}$ be revealed at time N and let $p_k = \mathbb{E}[Y \mid \mathcal{F}_k]$ be the ideal sequential forecast, a bounded Doob martingale with $p_0 \in (0, 1)$ and $p_N = Y$. Write

$$M_N := \max_{0 \leq k \leq N} p_k, \qquad M := \sup_{t \in [0,1]} p_t$$

for the discrete- and continuous-time path maxima.

Definition 1 (First-passage time)**.** For $x \in (0,1]$, define

$$\tau_x := \inf\{k \in \{0,1,\dots,N\} : p_k \geq x\},$$

with $\inf \varnothing := N+1$.

Our primary benchmark is the *peak-on-loss* law: the distribution of the maximum along realizations with $Y = 0$.

Theorem 1 (Peak-on-loss benchmark)**.** Under sequential calibration and continuous sample paths,

$$\mathbb{P}(M \geq x \mid Y = 0) = \frac{p_0}{1-p_0} \cdot \frac{1-x}{x}, \qquad x \in [p_0, 1).$$

Equivalently, for $x \in [p_0, 1)$,

$$F_{M|Y=0}(x) = 1 - \frac{p_0}{1-p_0} \cdot \frac{1-x}{x},$$

with $F_{M|Y=0}(x) = 0$ for $x < p_0$ and $F_{M|Y=0}(1) = 1$.

Theorem 2 (Discrete-time peak-on-loss bound)**.** In discrete time,

$$F_{M_N|Y=0}(x) \;\geq\; 1 - \frac{p_0}{1-p_0} \cdot \frac{1-x}{x}, \qquad x \in [p_0, 1),$$

with equality in the absence of terminal-step non-attainment and first-passage overshoots.

Remark 1 (Discrete-time correction identity)**.** For $x \in (p_0, 1)$,

$$\mathbb{P}(M_N \geq x \mid Y = 0) = \frac{p_0}{1-p_0} \cdot \frac{1-x}{x} \;-\; C_1(x) \;-\; C_2(x),$$

where

$$C_1(x) := \frac{1}{x}\,\mathbb{E}\big[(p_{\tau_x} - x)\,\mathbf{1}\{\tau_x \leq N\} \,\big|\, Y = 0\big], \qquad C_2(x) := \mathbb{P}(\tau_x > N \mid Y = 0).$$

Here C_1 captures first-passage overshoots and C_2 captures discrete-time non-attainment; both are nonnegative and vanish in settings without overshoots or terminal-step crossings (e.g. in the continuous-sample-path case, where the benchmark becomes an identity).

Full statements and proofs appear in Appendices A and B, and derivations for the competitive corollaries appear in Appendices C and D.

4 Extreme-path calibration diagnostics

We now translate the laws above into diagnostics. The null is the Doob-martingale calibration condition $p_k = \mathbb{P}(Y = 1 \mid \mathcal{F}_k)$. Departures can reflect update misspecification (drift, over-reaction/underreaction), filtration mismatch, or publication artifacts (discretization, rounding, smoothing, latency).

The same recipe applies to any extreme-path functional $T = T\big((p_k)_{k \leq N}, Y\big)$ with a closed-form conditional CDF (or a conservative discrete-time substitute). We present it for peak-on-loss, then aggregate across many paths.

4.1 Single binary outcome: peak-on-loss p-values

On $\{Y = 0\}$, the peak-on-loss statistic is $M_N = \max_{k \le N} p_k$. Under the continuous-path benchmark (Theorem 1), the one-sided tail probability for an observed peak m is

$$\mathbb{P}(M \ge m \mid Y = 0) \;=\; \left(\frac{p_0}{1-p_0}\right)\left(\frac{1-m}{m}\right), \qquad m \in [p_0, 1). \tag{1}$$

For discretely updated forecasts, (1) remains valid but conservative by Theorem 2.

4.2 Aggregating across many forecast paths

Suppose we observe independent realizations indexed by $i = 1, \ldots, n$. For each realization, compute an extreme-path statistic T_i (possibly conditional on the terminal outcome) together with an associated parameter θ_i (typically the initial probability value). Let $F_T(\cdot;\theta)$ denote the corresponding conditional CDF under the null (or a conservative substitute). Define

$$U_i := F_T(T_i; \theta_i).$$

Under the continuous-path law, each U_i is $\mathrm{Unif}(0,1)$, even when θ_i varies across realizations. With conservative F_T (as in discrete time), the upper tail of U_i is conservative: for $\alpha \in (0,1)$,

$$\mathbb{P}(U_i \ge 1 - \alpha) \le \alpha.$$

Equivalently, with $P_i := 1 - U_i$, the $\{P_i\}$ are super-uniform.

To test for upper-tail inflation (excess large U_i), we use the one-sided Kolmogorov–Smirnov statistic

$$D_n^{\text{upper}} := \sup_{0 \le t \le 1} \{t - \hat{F}_U(t)\},$$

where $\hat{F}_U$ is the empirical CDF of $\{U_i\}_{i=1}^n$. For the opposite direction (too few large U_i values), we use the mirror statistic

$$D_n^{\text{lower}} := \sup_{0 \le t \le 1} \{\hat{F}_U(t) - t\}.$$

Under the reparameterization $P_i = 1 - U_i$, these are equivalent to $D_n^- = \sup_t\{\hat{F}_P(t) - t\} = D_n^{\text{upper}}$ and $D_n^+ = \sup_t\{t - \hat{F}_P(t)\} = D_n^{\text{lower}}$.

5 Corollaries for competitive win-probability forecasts

The remaining objects are corollaries of the core peak-on-loss law. In competitive settings, "extreme probability assigned to the eventual loser" (or "extreme doubt assigned to the eventual winner") is a conditional extreme of a martingale or its complement, so the binary law transfers by conditioning and mixing over outcomes.

5.1 Loser's peak win probability in two-team games

Consider a two-team contest and encode the terminal outcome as $Y \in \{0,1\}$ where $Y = 1$ means Team A wins and $Y = 0$ means Team B wins. Let

$$p_t = \mathbb{P}(Y = 1 \mid \mathcal{F}_t) \qquad \text{and} \qquad q_t = 1 - p_t = \mathbb{P}(Y = 0 \mid \mathcal{F}_t)$$

denote the win-probability martingales for Teams A and B. The *loser's peak win probability* is the maximum win probability attained by the team that ultimately loses:

$$M_\lambda := \sup_{t\in[0,1]} \Big(p_t\,\mathbf{1}\{Y=0\} + q_t\,\mathbf{1}\{Y=1\}\Big).$$

Equivalently, on $\{Y=0\}$ the loser is Team A and $M_\lambda = \sup_t p_t$, while on $\{Y=1\}$ the loser is Team B and $M_\lambda = \sup_t q_t$. Therefore the distribution of M_λ is obtained by applying Theorem 1 to p_t on $\{Y=0\}$ and to the complement martingale q_t on $\{Y=1\}$, and then mixing over Y (details in Appendix C).

Assume without loss of generality that Team A is the pre-game favorite, so $p_0 \ge 1/2$. Under continuous sample paths, the benchmark CDF is

$$F_{M_\lambda}(x) = \begin{cases} 0 & x < 1-p_0, \\ 1 - \dfrac{1-p_0}{x} & 1-p_0 \le x < p_0, \\ 2 - \dfrac{1}{x} & p_0 \le x < 1, \\ 1 & x = 1. \end{cases}$$

The three regimes reflect which team can contribute at threshold x. For $x < 1-p_0$, neither team contributes; for $1-p_0 \le x < p_0$, only the underdog-as-loser contributes; for $x \ge p_0$, both contribute and the tail is *universal*:

$$\mathbb{P}(M_\lambda \ge x) = \frac{1}{x} - 1, \qquad x \in [p_0, 1).$$

In particular, when $p_0 \le 0.9$, $\mathbb{P}(M_\lambda \ge 0.9) = 1/9 \approx 11\%$. When $p_0 \ne 1/2$, the intermediate region $[1-p_0, p_0)$ produces a kink (Figure 1).

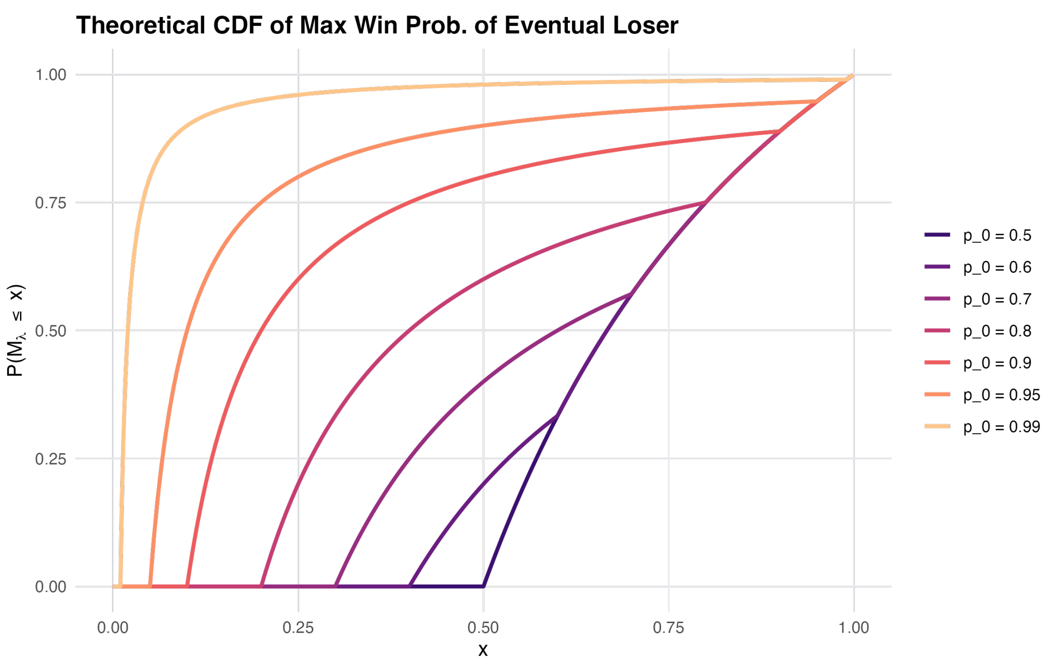


Figure 1: Benchmark CDF $F_{M_\lambda}(x)$ for different pre-game favorite probabilities p_0.

5.2 Winner's trough in n-team contests

Now consider $n \geq 3$ teams with exactly one winner. Let $Y^{(i)} = \mathbf{1}\{\text{team } i \text{ wins}\}$ and let

$$p_t^{(i)} = \mathbb{P}(Y^{(i)} = 1 \mid \mathcal{F}_t), \qquad i = 1, \dots, n,$$

be the corresponding win-probability martingales, with $\sum_{i=1}^n p_t^{(i)} = 1$. If W denotes the eventual winner, the *winner's trough* is simply the minimum probability assigned to the winner along the path:

$$M_\omega := \inf_{t\in[0,1]} p_t^{(W)} \;=\; \inf_{t\in[0,1]} \sum_{i=1}^n p_t^{(i)} \mathbf{1}\{Y^{(i)} = 1\}.$$

Its benchmark distribution follows by applying Theorem 1 to the complement martingale $1-p_t^{(i)} = \mathbb{P}(Y^{(i)} = 0 \mid \mathcal{F}_t)$ on the event $\{Y^{(i)} = 1\}$ and then mixing over i (derivation in Appendix D).

In the symmetric case $p_0^{(i)} = 1/n$, we obtain the closed form

$$F_{M_\omega}(x) = \frac{(n-1)x}{1-x}, \qquad x \in \Big[0, \frac{1}{n}\Big),$$

with $F_{M_\omega}(x) = 1$ for $x \geq 1/n$. For example, when $n = 3$, $\mathbb{P}(M_\omega \leq 0.2) = 0.5$. Figure 2 plots F_{M_ω} for several n.

More generally, when the pre-game probabilities $\{p_0^{(i)}\}$ are not equal, a convenient expression is

$$F_{M_\omega}(x) = \sum_{i=1}^n p_0^{(i)} \, \min\left\{1,\; \left(\frac{1-p_0^{(i)}}{p_0^{(i)}}\right)\left(\frac{x}{1-x}\right)\right\}, \qquad x \in [0,1),$$

which reduces to the symmetric formula above when $p_0^{(i)} \equiv 1/n$.

5.3 Interpretation

These corollaries calibrate common retrospective statements in competitive forecasting. For example, "the eventual loser was assigned 90%" can have substantial null probability; similarly, the winner's trough quantifies how low the eventual winner's probability can plausibly fall. Additional benchmark values for common thresholds are collected in Appendix E. Section 7 aggregates these game-level quantities through PIT diagnostics.

6 Simulation study

This section has three goals: (i) verify null calibration under K refinement, (ii) quantify discrete-time conservatism under feed coarsening, and (iii) map structural alternatives to directional PIT signatures.

6.1 Design

For each trajectory i, we draw $p_{0,i} \sim \text{Unif}(0.25, 0.75)$ and $Y_i \sim \text{Bernoulli}(p_{0,i})$. Let $B_i^0(t)$ be a standard Brownian bridge on $[0,1]$, independent of Y_i, and define the latent signal

$$X_i(t) = tY_i + \sigma B_i^0(t), \qquad t \in [0,1], \quad \sigma = 0.5.$$

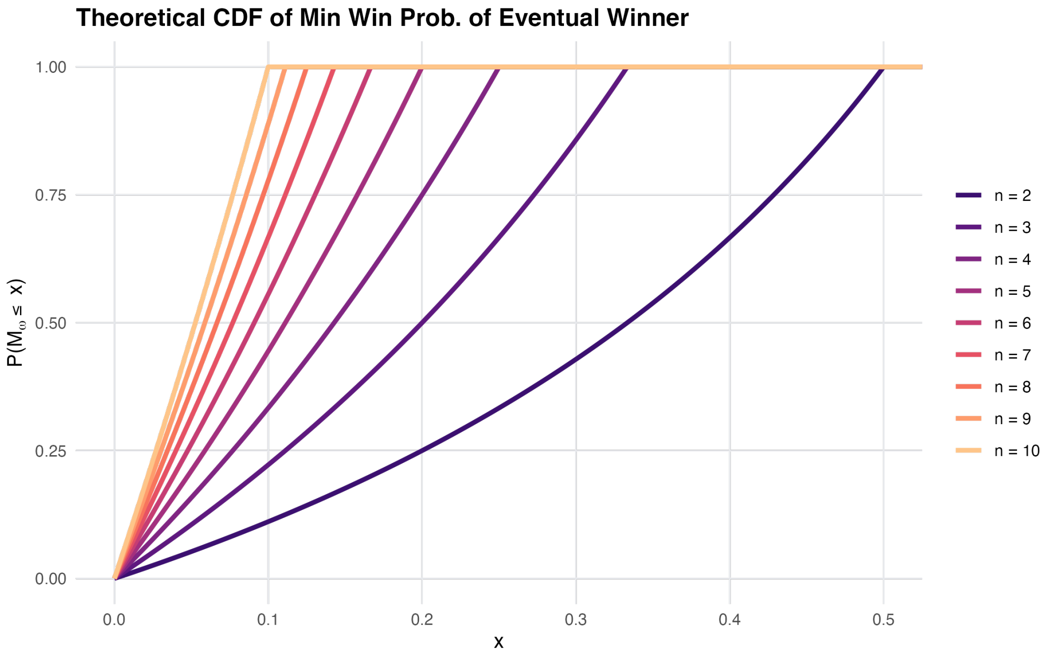


Figure 2: Benchmark CDF $F_{M_\omega}(x)$ for symmetric n-team games.

Because $B_i^0(1) = 0$, we have $X_i(1) = Y_i$. For $t < 1$, the posterior log-odds process is

$$\ell_i(t) = \mathrm{logit}(p_{0,i}) + \frac{X_i(t) - t/2}{\sigma^2(1-t)}, \qquad p_i(t) = \mathrm{logit}^{-1}\big(\ell_i(t)\big),$$

and we set $p_i(1) := Y_i$. This gives an anchored latent null with continuous paths on $[0, 1)$ and terminal revelation at $t = 1$. For the main discrete-feed experiments (coarsening and structural alternatives), we use

$$t_k^{\mathrm{base}} = \frac{k}{K_{\mathrm{base}}}, \quad k = 0, \dots, K_{\mathrm{base}}, \quad K_{\mathrm{base}} = 1000.$$

For null calibration, we separately refine K over $\{100, 1000, 10000, 100000, 1000000\}$ at fixed $n = 10000$.

To study departures from the ideal process, we map each latent path $p_i(t_k)$ to a reported path $\hat{p}_i(t_k)$ on a generic grid $k = 0, \dots, K$ using the following operators:

- **Over/underreaction:** $\hat{\ell}_i(t_k) = \beta \ell_i(t_k)$.
- **Predictable drift:** $\hat{\ell}_i(t_k) = \ell_i(t_k) + \delta\, t_k \tanh(\ell_i(t_k)/c)$ with $c = 2$.
- **Filtration lag:** $\hat{p}_i(t_k) = p_i(r_\eta(t_k))$, where $r_\eta(t) = t/(1 + \eta(1-t))$ and η indexes lag intensity: $\eta = 0$ is no lag (null), and larger η implies greater lag.
- **Smoothing:** $\hat{p}_{i0} = p_{0,i}$, $\hat{p}_{ik} = \lambda \hat{p}_{i,k-1} + (1-\lambda) p_i(t_k)$ for $k = 1, \dots, K-1$, and $\hat{p}_{iK} = Y_i$.
- **Feed coarsening and rounding:** keep every mth update and optionally round to a fixed grid (e.g., 0.01 or 0.05).

For each simulated dataset, we compute peak-on-loss PIT values U_i and summarize them with upper-tail frequencies, D_n^{upper}, and D_n^{lower}.

6.2 Null calibration via K refinement

Under the true continuous-process null, PIT values are uniform by Theorem 1. Figure 3 evaluates this calibration by plotting the null signature $t - \hat{F}_U(t)$ over an order-of-magnitude refinement range $K \in \{100, 1000, 10000, 100000, 1000000\}$ with fixed $n = 10000$. As K increases, the curves move toward the reference line at zero, matching the expected convergence to the continuous-path benchmark.

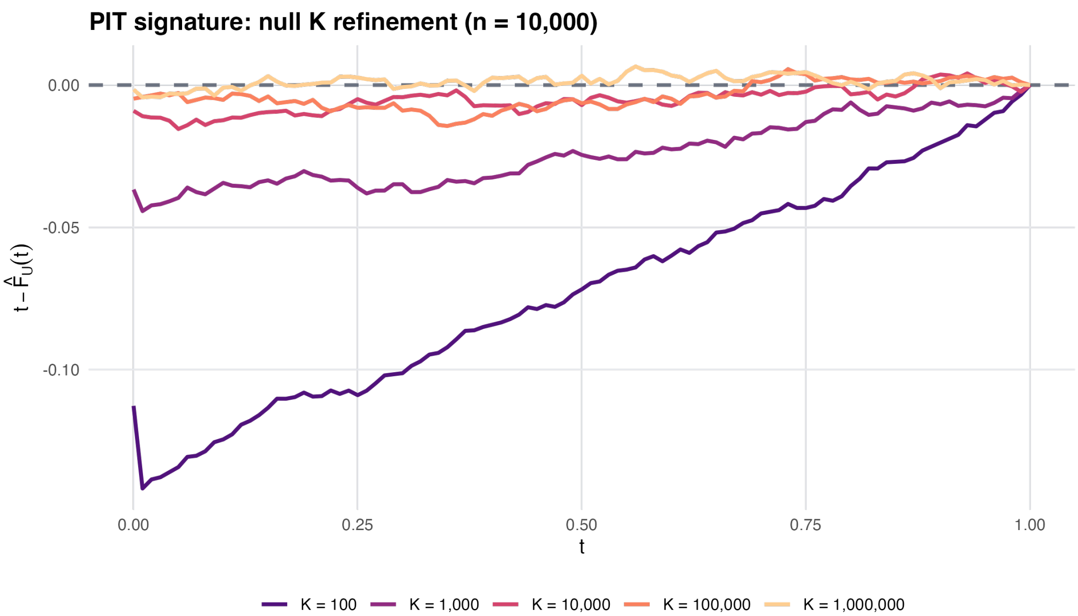


Figure 3: Null calibration under K refinement at fixed sample size $n = 10000$: the signature $t - \hat{F}_U(t)$ approaches the zero reference line as the feed is refined.

6.3 Discrete-time conservatism under coarsening

Figure 4 fixes a calibrated base process ($K_{\text{base}} = 1000$) and coarsens the observed feed by keeping every mth update. As coarsening increases, the signature $t - \hat{F}_U(t)$ shifts below zero, indicating conservative upper tails under discrete observation. The upper-tail frequencies $\widehat{\mathbb{P}}(U \geq u)$ decline with coarsening for $u \in \{0.90, 0.95, 0.99\}$, consistent with Theorem 2.

6.4 Structural alternatives and directional signatures

Figure 5 plots $t - \hat{F}_U(t)$ under representative alternatives. Overreaction and positive drift produce upward deviations (excess large U_i), while underreaction, smoothing, and filtration lag produce the opposite directional pattern. Curves can start slightly below zero near $t = 0$ because discrete/coarsened observation induces mild null conservatism in the lower tail (so $\hat{F}_U(t) > t$ for very small t). Coarsening mainly shifts the curve toward conservatism without mimicking the overreaction signature.

Table 1 reports estimated rejection rates at level 0.05 for both D_n^{upper} and D_n^{lower}. To account for discrete-time conservatism, the critical values are calibrated from repeated null draws at each sample size. The table uses path sample sizes $n \in \{1000, 5000\}$, with 120 Monte Carlo replicates at

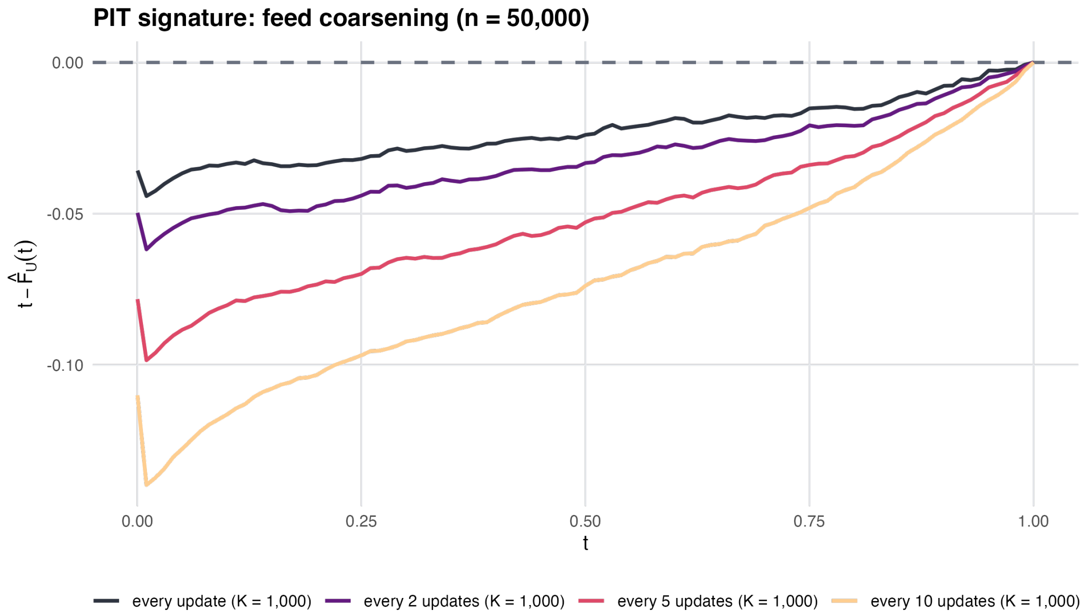


Figure 4: Null PIT signatures across coarsened feed resolutions from a common base process ($K_{\text{base}} = 1000$). Coarsening induces conservative upper tails.

$n = 1000$ and 100 replicates at $n = 5000$. Because conservative validity is directional in the upper-tail statistic, we treat D_n^{upper} as the primary inferential target; D_n^{lower} is reported as a directional sensitivity diagnostic. The resulting profile remains interpretable: D_n^{upper} is most sensitive to overreaction/drift (e.g., $\beta = 1.1$: 0.475 at $n = 1000$, 1.000 at $n = 5000$), while D_n^{lower} is strongest for smoothing and underreaction (e.g., $\lambda = 0.8$: 1.000 at both sizes; $\beta = 0.9$: 0.400 at $n = 1000$, 0.680 at $n = 5000$), and can also rise for strong overreaction at larger n. A broader stress-test grid with additional parameter levels and reporting-artifact variants appears in Appendix Table 4.

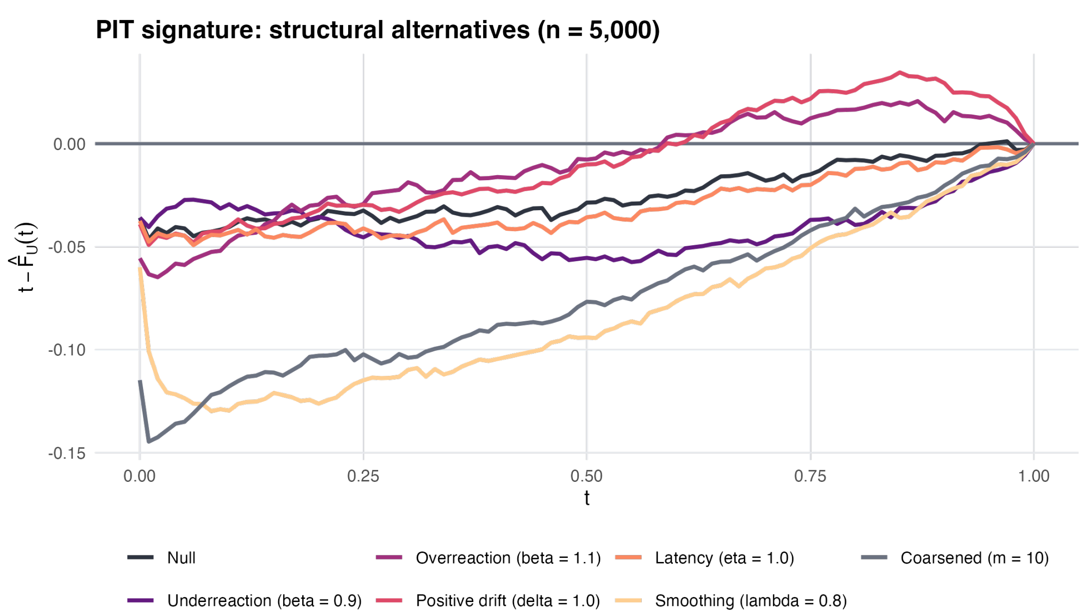


Figure 5: Directional PIT signatures under representative deviations from the martingale ideal.

Table 1: Rejection rates for D_n^{upper} and D_n^{lower} under structural alternatives ($\alpha = 0.05$).

Scenario	D_n^{upper} ($n = 1000$)	D_n^{upper} ($n = 5000$)	D_n^{lower} ($n = 1000$)	D_n^{lower} ($n = 5000$)
Null	0.050	0.050	0.050	0.050
$\beta = 0.9$	0.000	0.000	0.400	0.680
$\beta = 1.1$	0.475	1.000	0.142	0.970
$\delta = 0.5$	0.300	0.950	0.058	0.070
$\delta = 1.0$	0.750	1.000	0.042	0.090
$\eta = 1.0$	0.033	0.020	0.150	0.240
$\lambda = 0.5$	0.000	0.000	0.733	1.000
$\lambda = 0.8$	0.000	0.000	1.000	1.000

7 Empirical study: ESPN win-probability models in the NFL and NBA

This section illustrates the diagnostics on a widely visible forecasting system: live ESPN win probabilities. The aim is methodological: test whether observed path extremes match the null predictions.

7.1 Data sources and preprocessing

We analyze ESPN play-by-play win-probability data for NFL and NBA regular-season games from 2018–2024 (ESPN, 2025). NFL data were obtained via `espnscrapeR` (Mock, 2025), and NBA data via `hoopR` (Gilani, 2023). After excluding ties and games with missing scores or win-probability series, we retain $n = 1832$ NFL games and $n = 8261$ NBA games. For each game, we define p_0 as the maximum of the two teams' first-reported win probabilities (so that $p_0 \geq 0.5$) and compute M_λ as the maximum win probability attained by the eventual loser over the game (Section 5.1).

7.2 Null comparison via probability integral transform

Because the benchmark distribution of M_λ depends on p_0, we aggregate across games using a probability integral transform. For game i, let $p_{0,i} \geq 1/2$ be the initial reported favorite probability and let $m_i = M_{\lambda,i}$ be the observed eventual-loser peak. Define

$$U_i := F_{M_\lambda}(m_i; p_{0,i}).$$

Under the continuous-path law, $\{U_i\}$ are i.i.d. Unif$(0, 1)$. For discretely updated feeds, the upper tail of $\{U_i\}$ is conservative (as predicted by Theorem 2 and confirmed in Section 6), so excess large values indicate overly extreme eventual-loser peaks. We use U_i as the main display scale and note the equivalent reparameterization $P_i = 1 - U_i$.

7.3 Global test and tail summaries

Our primary global diagnostic is upper-tail inflation. Let $\hat{F}_U$ be the empirical CDF of $\{U_i\}_{i=1}^n$ and define

$$D_n^{\text{upper}} := \sup_{0 \leq t \leq 1} \{t - \hat{F}_U(t)\}.$$

Large D_n^{upper} indicates excess large U_i values; this direction remains valid under discrete-time conservatism.

As descriptive complements, we report $\widehat{\mathbb{P}}_n(U \geq u)$ for $u \in \{0.90, 0.95, 0.99\}$; values substantially above $\{0.10, 0.05, 0.01\}$ indicate upper-tail inflation.

Figure 6 plots ECDFs of $\{U_i\}$ against the uniform-reference line $t \mapsto t$ for both leagues. Tables 2 and 3 report D_n^{upper}, its one-sided p-value, and the tail summaries.

Table 2: NFL PIT diagnostic summary.

n	$\widehat{\mathbb{P}}_n(U \geq 0.90)$	$\widehat{\mathbb{P}}_n(U \geq 0.95)$	$\widehat{\mathbb{P}}_n(U \geq 0.99)$	D_n^{upper}	p-value
1832	0.086	0.046	0.017	0.0153	0.424

Table 3: NBA PIT diagnostic summary.

n	$\widehat{\mathbb{P}}_n(U \geq 0.90)$	$\widehat{\mathbb{P}}_n(U \geq 0.95)$	$\widehat{\mathbb{P}}_n(U \geq 0.99)$	D_n^{upper}	p-value
8261	0.131	0.070	0.020	0.0962	<0.0001

7.4 Interpretation of the diagnostic tests

The NFL results (Table 2) show no strong evidence against the null in the tested direction. The empirical upper-tail frequencies at $u \in \{0.90, 0.95\}$ are close to nominal exceedance rates (0.086 vs. 0.10, 0.046 vs. 0.05), while the $u = 0.99$ tail is mildly elevated (0.017 vs. 0.01). The one-sided upper-tail K–S statistic does not reject ($p = 0.42$), indicating no systematic upper-tail inflation.

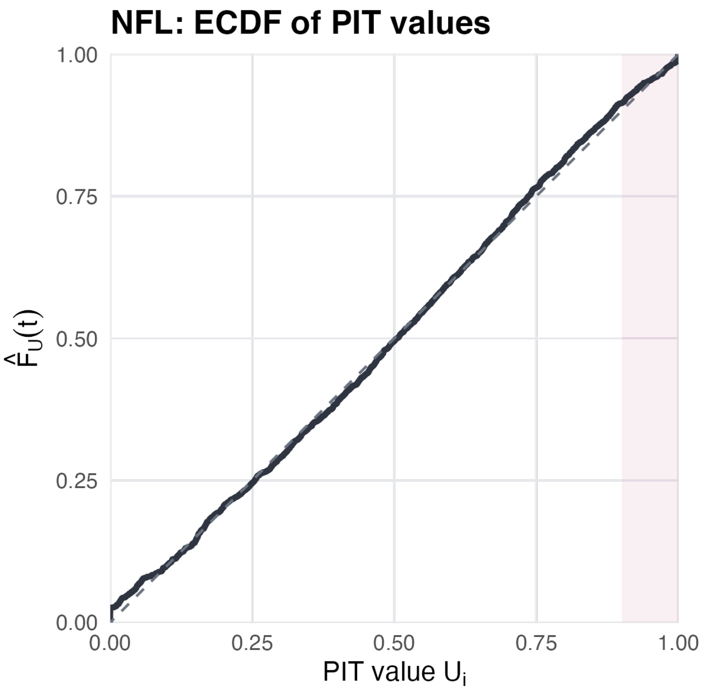
The NBA results (Table 3) show clear departures. Upper-tail mass is inflated (0.131 vs. 0.10, 0.070 vs. 0.05, 0.020 vs. 0.01), and the one-sided K–S statistic is large ($p < 10^{-4}$). We therefore reject the null in the direction of excessively high eventual-loser peaks. This discrepancy is consistent with overconfident or overly reactive updating, but could also reflect filtration mismatch or reporting/estimation artifacts in the published series (Section 8).

8 Discussion

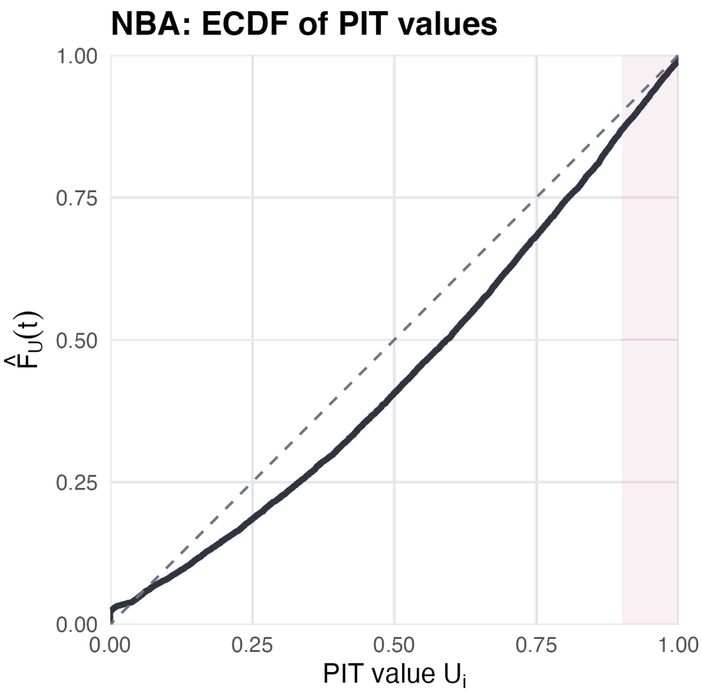
Narratives around sequential probability forecasts often emphasize retrospective extremes: "the team reached 90% and still lost." The relevant baseline for these statements is the *conditional* law of path extrema given the terminal outcome, rather than fixed-time heuristics such as $1 - m$. This perspective turns narratives about "peak probabilities" into testable distributional claims. In settings such as sports win probabilities, departures may reflect update misspecification, filtration mismatch, or publication artifacts rather than the intrinsic surprisingness of any single collapse.

8.1 Limitations and assumptions

Our theoretical guarantees are statements about the ideal process $p_k = \mathbb{E}[Y \mid \mathcal{F}_k]$. In practice, reported win probabilities estimate this conditional expectation, so empirical discrepancies may also reflect model misspecification, nonstationarity, or information-set mismatch (the model's effective information set not aligning with the outcome-relevant filtration).



(a) NFL



(b) NBA

Figure 6: ECDFs of PIT values U_i with the uniform-reference line $t \mapsto t$.

Exact identities additionally require continuous sample paths, which rules out terminal-step crossings and first-passage overshoots. For discretely updated feeds, our discrete-time results remain valid but can be conservative; the gap is governed by explicit correction terms associated with overshoots and non-attainment. Finally, we treat binary terminal outcomes $p_N \in \{0, 1\}$; extensions to ties or other non-binary terminal structures require different arguments.

8.2 Extensions

Several extensions are natural. First, relax the terminal binary condition to allow $p_N \in [0, 1]$ (e.g. ties or partial-credit outcomes). Second, derive analogous laws for other path functionals, such as peak timing, joint laws like $(\sup_t p_t, p_1)$, or multistream extremes. Third, extend the structured-deviation analysis beyond the one-parameter families studied here (sensitivity scaling, predictable drift, filtration lag, and smoothing) to include richer jump components and multistream interactions. Together, these directions preserve the core perspective: conditional extreme-value laws provide a simple, model-agnostic way to evaluate sequential probability forecasts using the same peak-probability framing used in practice while retaining a principled calibration interpretation.

Appendix

A Proofs of Discrete-Time Results

Throughout, $(p_k)_{k=0}^N$ is a bounded martingale with $p_0 \in (0,1)$ and terminal value $p_N = Y \in \{0,1\}$. For $x \in (0,1]$, define the first-passage time

$$\tau_x := \inf\{k \in \{0,1,\dots,N\} : p_k \geq x\},$$

with $\inf \varnothing := N+1$.

A.1 Unconditional distribution of M_N

Theorem 3 (Discrete-time unconditional bound)**.** Let $M_N = \max_{0\leq k\leq N} p_k$. For $x \in [p_0, 1)$,

$$F_{M_N}(x) = \mathbb{P}(M_N \leq x) \;\geq\; 1 - \frac{p_0}{x},$$

with equality if $\mathbb{P}(\tau_x = N) = 0$ and $p_{\tau_x} = x$ a.s. on $\{\tau_x < N\}$. For $x < p_0$, $F_{M_N}(x) = 0$. Moreover, M_N has an atom at 1 with $\mathbb{P}(M_N = 1) = p_0$.

Proof. If $x < p_0$ then $M_N \geq p_0 > x$ a.s., hence $F_{M_N}(x) = 0$.

Fix $x \in [p_0, 1)$. By optional stopping for the bounded martingale (p_k) at $\tau_x \wedge N$,

$$\mathbb{E}[p_{\tau_x \wedge N}] = p_0.$$

Decompose $p_{\tau_x\wedge N} = p_{\tau_x}\mathbf{1}\{\tau_x < N\} + p_N\mathbf{1}\{\tau_x \geq N\}$. On $\{\tau_x < N\}$ we have $p_{\tau_x} \geq x$. Also $p_N = Y \in \{0,1\}$ and $\mathbf{1}\{\tau_x \geq N\} \leq 1$. Therefore,

$$p_0 = \mathbb{E}[p_{\tau_x}\mathbf{1}\{\tau_x < N\}] + \mathbb{E}[p_N\mathbf{1}\{\tau_x \geq N\}] \;\geq\; x\,\mathbb{P}(\tau_x < N).$$

Since $\{\tau_x < N\} = \{M_N \geq x\}$ for $x < 1$, this yields

$$\mathbb{P}(M_N \geq x) \leq \frac{p_0}{x} \qquad \Rightarrow \qquad F_{M_N}(x) = 1 - \mathbb{P}(M_N > x) \geq 1 - \frac{p_0}{x}.$$

Equality holds when (i) there is no terminal-step crossing $\mathbb{P}(\tau_x = N) = 0$ and (ii) there is no overshoot $p_{\tau_x} = x$ a.s. on $\{\tau_x < N\}$.

Finally, $M_N = 1$ iff $p_N = 1$ iff $Y = 1$, so $\mathbb{P}(M_N = 1) = p_0$. □

A.2 Conditional distribution of M_N given $Y = 0$

Theorem 4 (Discrete-time conditional bound)**.** For $x \in [p_0, 1)$,

$$F_{M_N|Y=0}(x) = \mathbb{P}(M_N \leq x \mid Y = 0) \;\geq\; 1 - \Big(\frac{p_0}{1-p_0}\Big)\Big(\frac{1-x}{x}\Big),$$

with equality under the same no-crossing/no-overshoot conditions as in Theorem 3. For $x < p_0$, $F_{M_N|Y=0}(x) = 0$.

Proof. Fix $x \in [p_0, 1)$. Decompose $\mathbb{P}(M_N \geq x)$ by Y:

$$\mathbb{P}(M_N \geq x) = \mathbb{P}(M_N \geq x, Y = 1) + \mathbb{P}(M_N \geq x, Y = 0).$$

On $\{Y = 1\}$ we have $p_N = 1 \geq x$, hence $\{Y = 1\} \subseteq \{M_N \geq x\}$ and

$$\mathbb{P}(M_N \geq x) = \mathbb{P}(Y = 1) + \mathbb{P}(Y = 0)\mathbb{P}(M_N \geq x \mid Y = 0) = p_0 + (1 - p_0)\mathbb{P}(M_N \geq x \mid Y = 0).$$

By Theorem 3, $\mathbb{P}(M_N \geq x) \leq p_0/x$, so

$$\frac{p_0}{x} \;\geq\; p_0 + (1 - p_0)\mathbb{P}(M_N \geq x \mid Y = 0),$$

which rearranges to

$$\mathbb{P}(M_N \geq x \mid Y = 0) \;\leq\; \Big(\frac{p_0}{1 - p_0}\Big)\Big(\frac{1 - x}{x}\Big).$$

Taking complements gives the stated lower bound on $F_{M_N|Y=0}(x)$. $\square$

B Proofs of Continuous-Path Results

Let $(p_t)_{t\in[0,1]}$ be a bounded martingale with $p_0 \in (0, 1)$ and $p_1 = Y \in \{0, 1\}$, and assume path continuity. Define $M = \sup_{t\in[0,1]} p_t$ and $\tau_x = \inf\{t \in [0, 1] : p_t \geq x\}$.

B.1 Unconditional distribution of M

Theorem 5 (Continuous-path unconditional)**.** For $x \in [p_0, 1)$,

$$F_M(x) = \mathbb{P}(M \leq x) = 1 - \frac{p_0}{x},$$

and $F_M(x) = 0$ for $x < p_0$, while $F_M(1) = 1$. In particular, $\mathbb{P}(M = 1) = p_0$.

Proof. If $x < p_0$ then $M \geq p_0 > x$ a.s.

Fix $x \in [p_0, 1)$. Optional stopping at $\tau_x \wedge 1$ yields $\mathbb{E}[p_{\tau_x\wedge 1}] = p_0$. By continuity, on $\{\tau_x < 1\}$ we have $p_{\tau_x} = x$, and on $\{\tau_x \geq 1\}$ we have $p_1 = Y$. Thus

$$p_0 = x\,\mathbb{P}(\tau_x < 1) + \mathbb{E}[Y\mathbf{1}\{\tau_x \geq 1\}].$$

But if $\tau_x \geq 1$ and $Y = 1$, then $p_1 = 1 \geq x$ forces $\tau_x \leq 1$, a contradiction; hence $\mathbb{E}[Y\mathbf{1}\{\tau_x \geq 1\}] = 0$ for $x < 1$. Therefore $p_0 = x\,\mathbb{P}(\tau_x < 1)$, i.e. $\mathbb{P}(M \geq x) = \mathbb{P}(\tau_x < 1) = p_0/x$. The CDF follows by complement. Finally, $M = 1$ iff $Y = 1$, so $\mathbb{P}(M = 1) = p_0$. $\square$

B.2 Conditional distribution of M given $Y = 0$

Theorem 6 (Continuous-path conditional)**.** For $x \in [p_0, 1)$,

$$F_{M|Y=0}(x) = \mathbb{P}(M \leq x \mid Y = 0) = 1 - \Big(\frac{p_0}{1 - p_0}\Big)\Big(\frac{1 - x}{x}\Big),$$

and $F_{M|Y=0}(x) = 0$ for $x < p_0$, while $F_{M|Y=0}(1) = 1$.

Proof. Fix $x \in [p_0, 1)$. As before,

$$\mathbb{P}(M \geq x) = \mathbb{P}(Y = 1) + (1 - p_0)\mathbb{P}(M \geq x \mid Y = 0) = p_0 + (1 - p_0)\mathbb{P}(M \geq x \mid Y = 0).$$

By Theorem 5, $\mathbb{P}(M \geq x) = p_0/x$, hence

$$\frac{p_0}{x} = p_0 + (1 - p_0)\mathbb{P}(M \geq x \mid Y = 0) \quad \Rightarrow \quad \mathbb{P}(M \geq x \mid Y = 0) = \left(\frac{p_0}{1 - p_0}\right)\left(\frac{1 - x}{x}\right).$$

Taking complements yields the CDF. Finally, on $\{Y = 0\}$ the process cannot attain 1 (if $p_t = 1$ then $\mathbb{P}(Y = 1 \mid \mathcal{F}_t) = 1$), so $M < 1$ a.s. and $F_{M|Y=0}(1) = 1$. $\square$

C Distribution of M_λ (Two Teams)

Recall the two-team setting with win-probability martingales (p_t) and (q_t), where $q_t = 1 - p_t$ and $p_0 = \mathbb{P}(Y = 1)$; assume $p_0 \geq 0.5$ without loss of generality. The maximum win probability attained by the eventual loser is

$$M_\lambda \equiv \sup_{0 \leq t \leq 1} \Big(p_t \mathbf{1}\{Y = 0\} + q_t \mathbf{1}\{Y = 1\}\Big). \tag{2}$$

By the law of total probability,

$$F_{M_\lambda}(x) = \mathbb{P}(M_\lambda \leq x) = \mathbb{P}(M_\lambda \leq x \mid Y = 0)(1 - p_0) + \mathbb{P}(M_\lambda \leq x \mid Y = 1)p_0. \tag{3}$$

From the conditional distributions of the previous section, when team A loses we have

$$F_{M_\lambda|Y=0}(x) = \mathbb{P}(M_\lambda \leq x \mid Y = 0) = \begin{cases} 0 & \text{for } x < p_0, \\ 1 - \left(\frac{p_0}{1-p_0}\right)\left(\frac{1-x}{x}\right) & \text{for } x \in [p_0, 1), \\ 1 & \text{for } x = 1. \end{cases} \tag{4}$$

Applying the same formula with p_0 replaced by $1 - p_0$ gives

$$F_{M_\lambda|Y=1}(x) = \mathbb{P}(M_\lambda \leq x \mid Y = 1) = \begin{cases} 0 & \text{for } x < 1 - p_0, \\ 1 - \left(\frac{1-p_0}{p_0}\right)\left(\frac{1-x}{x}\right) & \text{for } x \in [1 - p_0, 1), \\ 1 & \text{for } x = 1. \end{cases} \tag{5}$$

With $p_0 \geq 0.5$ (team A favored), substituting (4) and (5) into (3) gives three regions:

1. For $0 \leq x < 1 - p_0$, neither (4) nor (5) contributes, so $F_{M_\lambda}(x) = 0$.
2. For $1 - p_0 \leq x < p_0$, only (5) contributes:

$$\begin{aligned} F_{M_\lambda}(x) &= (1 - p_0) \cdot 0 + p_0 \cdot \left[1 - \left(\frac{1 - p_0}{p_0}\right)\left(\frac{1 - x}{x}\right)\right] \\ &= p_0 - (1 - p_0)\left(\frac{1 - x}{x}\right) = 1 - \frac{1 - p_0}{x}. \end{aligned}$$

3. For $x \ge p_0$, both (4) and (5) contribute:

$$\begin{aligned}F_{M_\lambda}(x) &= (1-p_0)\left[1-\left(\frac{p_0}{1-p_0}\right)\left(\frac{1-x}{x}\right)\right]+p_0\left[1-\left(\frac{1-p_0}{p_0}\right)\left(\frac{1-x}{x}\right)\right]\\ &= \left[(1-p_0)-p_0\left(\frac{1-x}{x}\right)\right]+\left[p_0-(1-p_0)\left(\frac{1-x}{x}\right)\right]\\ &= 1-\frac{1-x}{x}=\frac{2x-1}{x}=2-\frac{1}{x}.\end{aligned}$$

Summarizing, the piecewise cumulative distribution function is:

$$F_{M_\lambda}(x)=\mathbb{P}(M_\lambda \le x)=\begin{cases}0 & \text{for } 0 \le x < 1-p_0,\\ 1-\frac{1-p_0}{x} & \text{for } 1-p_0 \le x < p_0,\\ 2-\frac{1}{x} & \text{for } p_0 \le x < 1,\\ 1 & \text{for } x = 1.\end{cases}$$

D Distribution of M_ω (n Teams)

Consider an n-team game with win-probability martingales $(p_t^{(i)})_{0\le t\le 1}$ for $i \in \{1,\dots,n\}$, where $Y^{(i)} = \mathbf{1}\{\text{team } i \text{ wins}\}$ and $p_0^{(i)} = \mathbb{P}(Y^{(i)}=1)$. Define the eventual-winner minimum

$$M_\omega \equiv \inf_{0\le t\le 1} \sum_{i=1}^{n} p_t^{(i)} \mathbf{1}\{Y^{(i)}=1\}. \tag{6}$$

On the event $\{Y^{(i)}=1\}$, the winner's path is $(p_t^{(i)})$, so

$$M_\omega = \inf_{0\le t\le 1} p_t^{(i)} \quad \text{on } \{Y^{(i)}=1\}.$$

Since $\inf_t p_t^{(i)} \le x$ if and only if $\sup_t(1-p_t^{(i)}) \ge 1-x$, the conditional distribution follows from Theorem 6 applied to the complementary martingale $(1-p_t^{(i)}) = \mathbb{P}(Y^{(i)}=0 \mid \mathcal{F}_t)$ with initial value $1-p_0^{(i)}$:

$$\begin{aligned}\mathbb{P}(M_\omega \le x \mid Y^{(i)}=1) &= \mathbb{P}\left(\sup_{0\le t\le 1}(1-p_t^{(i)}) \ge 1-x \,\middle|\, Y^{(i)}=1\right)\\ &= \begin{cases}0 & \text{for } x<0,\\ \left(\frac{1-p_0^{(i)}}{p_0^{(i)}}\right)\left(\frac{x}{1-x}\right) & \text{for } x \in [0,p_0^{(i)}),\\ 1 & \text{for } x \ge p_0^{(i)}.\end{cases}\end{aligned} \tag{7}$$

Finally, by the law of total probability,

$$\begin{aligned}F_{M_\omega}(x)=\mathbb{P}(M_\omega \le x) &= \sum_{i=1}^{n}\mathbb{P}(M_\omega \le x \mid Y^{(i)}=1)\mathbb{P}(Y^{(i)}=1)\\ &= \sum_{i=1}^{n}\mathbb{P}(M_\omega \le x \mid Y^{(i)}=1)p_0^{(i)}.\end{aligned} \tag{8}$$

Substituting (7) into (8) yields the piecewise distribution reported in the main text:

$$F_{M_\omega}(x) = \mathbb{P}(M_\omega \le x) = \begin{cases} 0 & \text{for } x < 0, \\ \sum_{i: x \ge p_0^{(i)}} p_0^{(i)} + \left(\frac{x}{1-x}\right) \sum_{i: x < p_0^{(i)}} (1 - p_0^{(i)}) & \text{for } x \in [0, \max_i p_0^{(i)}), \\ 1 & \text{for } x \ge \max_i p_0^{(i)}. \end{cases} \tag{9}$$

E Numerical Benchmarks

Under the martingale ideal with continuous sample paths, the reported values are exact benchmark probabilities for the corresponding extreme events; in discrete time, the same expressions may be interpreted as conservative upper bounds on tail probabilities.

E.1 Loser's peak win probability (two teams)

Recall the eventual loser's peak M_λ (the maximum win probability attained by the team that ultimately loses). Throughout this subsection, p_0 denotes the *pre-game favorite*'s win probability (so $p_0 \ge 1/2$); equivalently, relabel teams if needed so that this holds. Its CDF is

$$F_{M_\lambda}(x) = \begin{cases} 0, & 0 \le x < 1 - p_0, \\ 1 - \dfrac{1-p_0}{x}, & 1 - p_0 \le x < p_0, \\ 2 - \dfrac{1}{x}, & p_0 \le x < 1, \\ 1, & x = 1. \end{cases}$$

In particular, for thresholds $x \ge p_0$ the tail probability is universal:

$$\mathbb{P}(M_\lambda \ge x) = \frac{1}{x} - 1, \qquad x \in [p_0, 1),$$

so the probability of outcomes described as "reached 90% and still lost" is *independent* of pre-game strength whenever the peak exceeds the favorite's initial probability.

Symmetric case ($p_0 = 0.5$). Here $F_{M_\lambda}(x) = 2 - \frac{1}{x}$ on $[0.5, 1)$, and

$$\mathbb{P}(M_\lambda \ge 2/3) = \frac{1}{2}, \qquad \mathbb{P}(M_\lambda \ge 3/4) = \frac{1}{3}, \qquad \mathbb{P}(M_\lambda \ge 0.9) = \frac{1}{9} \approx 0.111.$$

Asymmetric case ($p_0 = 0.75$). The CDF has an intermediate region on $[0.25, 0.75)$:

$$F_{M_\lambda}(x) = \begin{cases} 0, & 0 \le x < 0.25, \\ 1 - \dfrac{0.25}{x}, & 0.25 \le x < 0.75, \\ 2 - \dfrac{1}{x}, & 0.75 \le x < 1, \\ 1, & x = 1. \end{cases}$$

For thresholds $x \geq 0.75$, the same universal tail applies:

$$\mathbb{P}(M_\lambda \geq 3/4) = \frac{1}{3}, \qquad \mathbb{P}(M_\lambda \geq 0.9) = \frac{1}{9} \approx 0.111.$$

(For a sub-favorite threshold such as $x = 0.5 < p_0$, the tail depends on p_0; e.g. $\mathbb{P}(M_\lambda \geq 0.5) = 0.5$ here.)

E.2 Winner's trough (n teams)

Recall the eventual winner's trough M_ω (the minimum win probability attained by the eventual winner). In the symmetric n-team case with $p_0^{(i)} = 1/n$, the CDF is

$$F_{M_\omega}(x) = \begin{cases} 0, & x < 0, \\ \dfrac{(n-1)x}{1-x}, & 0 \leq x < 1/n, \\ 1, & x \geq 1/n. \end{cases}$$

Thus M_ω is supported on $[0, 1/n]$ and becomes more concentrated near 0 as n grows.

Symmetric three-team case ($n = 3$). For $x \in [0, 1/3)$, $F_{M_\omega}(x) = \frac{2x}{1-x}$, so

$$\mathbb{P}(M_\omega \leq 0.2) = \frac{1}{2}, \qquad \mathbb{P}(M_\omega \leq 0.1) = \frac{2}{9} \approx 0.222, \qquad \mathbb{P}(M_\omega \leq 0.05) = \frac{2}{19} \approx 0.105.$$

Asymmetric three-team case $(p_0^{(1)}, p_0^{(2)}, p_0^{(3)}) = (1/6, 1/3, 1/2)$. Using the general mixture formula for asymmetric n-team games, the CDF is

$$F_{M_\omega}(x) = \begin{cases} 2 \cdot \dfrac{x}{1-x}, & 0 \leq x < 1/6, \\ \dfrac{1}{6} + \dfrac{7}{6} \cdot \dfrac{x}{1-x}, & 1/6 \leq x < 1/3, \\ \dfrac{1}{2} + \dfrac{1}{2} \cdot \dfrac{x}{1-x}, & 1/3 \leq x < 1/2, \\ 1, & x \geq 1/2, \end{cases}$$

so, for example,

$$\mathbb{P}(M_\omega \leq 0.1) = \frac{2}{9} \approx 0.222, \qquad \mathbb{P}(M_\omega \leq 0.25) = \frac{5}{9} \approx 0.556, \qquad \mathbb{P}(M_\omega \leq 0.4) = \frac{5}{6} \approx 0.833.$$

F Extended Simulation Stress Tests

Table 4 reports a broader simulation grid than the compact main-text table. It adds extra over/underreaction levels ($\beta \in \{0.8, 0.9, 1.1, 1.2\}$), additional drift and smoothing strengths, and separate rounding-only, coarsening-only, and combined reporting-artifact stress tests. As in Section 6, rejection uses null-calibrated critical values by sample size at nominal level $\alpha = 0.05$. The full table below uses the same Monte Carlo design as the main table: 120 replicates at $n = 1000$ and 100 replicates at $n = 5000$.

Table 4: Full simulation-suite rejection rates for D_n^{upper} and D_n^{lower} ($\alpha = 0.05$).

Scenario	D_n^{upper} ($n = 1000$)	D_n^{upper} ($n = 5000$)	D_n^{lower} ($n = 1000$)	D_n^{lower} ($n = 5000$)
Null	0.050	0.050	0.050	0.050
$\beta = 0.8$	0.000	0.000	0.908	1.000
$\beta = 0.9$	0.000	0.000	0.400	0.680
$\beta = 1.1$	0.475	1.000	0.142	0.970
$\beta = 1.2$	0.908	1.000	0.692	1.000
$\delta = 0.25$	0.142	0.450	0.058	0.060
$\delta = 0.5$	0.300	0.950	0.058	0.070
$\delta = 1.0$	0.750	1.000	0.042	0.090
$\eta = 0.5$	0.033	0.020	0.183	0.390
$\eta = 1.0$	0.033	0.020	0.150	0.240
$\lambda = 0.3$	0.025	0.020	0.400	0.840
$\lambda = 0.5$	0.000	0.000	0.733	1.000
$\lambda = 0.8$	0.000	0.000	1.000	1.000
round 0.01	0.058	0.080	0.050	0.050
round 0.05	0.108	1.000	0.117	0.680
coarsen $m = 5$	0.000	0.000	0.983	1.000
coarsen $m = 10$	0.000	0.000	1.000	1.000
coarsen $m = 10$ + round 0.05	0.000	1.000	1.000	1.000

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